A Neuro-Symbolic Harness for the ChinaTravel Travel Planning Challenge Phase 2 Technical Report of Team "Antarctic penguins"

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Abstract

We describe the Phase-2 harness of team Antarctic penguins for the ChinaTravel Travel Planning Challenge [LAMDA-NeSy, 2026; Shao et al., 2024]. The system is neuro-symbolic: a served Qwen3.6-27B [Qwen Team, 2025] is used only for natural-language-to-DSL translation, and everything downstream — constraint canonicalization, symbolic search, and soft-metric enrichment — is deterministic and mechanically verified against the generated constraints, never the hidden oracle. Phase 1 finished at 100.000 on the public leaderboard. For Phase 2 we adapted the system to the substantially tightened official evaluator (per-day meal-type limits, cross-activity chronology, transport–position consistency, and pass-rate-coupled preference scoring) with three gated repair passes (deoverlap, fixspace, mustpoi); on the 100 familiarization queries under fully honest live-NL conditions (oracle fields stripped, real Qwen3.6-27B), these passes lift Overall from 80.36 to 93.54 (FPR $76 \to 93$), and disclosed targeted repair plus gated soft-metric uplift raise the familiarization set to **99.18** with FPR and C-LPR at 100. The harness answers the organizers' fixed entry command, honors the 5-hour/100-query budget with per-query caps and a deterministic fallback, survives oracle-free held-out data by construction, and was verified end-to-end against a protocol-exact SGLang simulator with zero environment overrides.

1 Results Summary

The scoring formula in both phases is

$$\begin{aligned}\text{Overall} = {} & 0.10\,\text{EPR}_{\text{mic}} + 0.10\,\text{EPR}_{\text{mac}} + 0.25\,\text{C-LPR} \\ & + 0.40\,\text{FPR} + 0.05\,(\text{DAV} + \text{ATT} + \text{DDR}).\end{aligned}$$

FPR's 0.40 weight dominates, and the newest evaluator counts failed plans as zero in the preference terms, so feasibility strictly precedes soft-metric optimization everywhere in our design.

Setting	Overall
Phase 1, official leaderboard (v13)	**100.000**
Phase 2, live-NL familiarization-100 (local)	**99.18**
Phase 2, organizer-run held-out	—

Table 1: Headline results. The live-NL figure uses the organizers' familiarization set, the newest public evaluation code, the real Qwen3.6-27B, and no oracle access of any kind at inference time; the held-out figure will be reported by the organizers.

2 System Architecture

The pipeline uses the LLM only where language understanding is irreplaceable; all downstream stages are deterministic:

1. NL→DSL translation (Qwen3.6-27B via the organizers' SGLang endpoint, thinking disabled, temperature 0, fixed seed). Prompts are hardened against failure modes measured on Qwen itself (e.g. a stock few-shot example that induced invented room constraints; fixing it alone moved measured translation fidelity from 46% to ∼89%). Mechanical verifiers and normalizers reject or repair malformed DSL with no oracle access.
2. DSL canonicalization: constraint literals, aliases, and quoting are normalized so the symbolic layer sees a stable language regardless of LLM phrasing variance.
3. Symbolic planning (UrbanTripOptimizedV6): phased bundle search over intercity transport, accommodation, attractions, and meals with geographic anchoring, DFS memoization, budget-drop repair, and a travel-day timing bias (rewarding early arrival / late departure; worth DAV +4.4, DDR +9.6 on fresh plans in Phase 1).
4. Gated enrichment battery (16 stages): every edit is kept only if the plan still passes schema, commonsense, and the generated hard constraints, and a soft metric strictly improves with none regressing. The gate never reads oracle fields — which the formal held-out data does not contain anyway.

Stage	EPR_{mic}	EPR_{mac}	C-LPR	FPR	DAV	ATT	DDR	Overall
live-NL baseline	99.37	84.00	82.69	76	74.88	74.42	69.72	80.36
+fixspace	99.93	99.00	95.49	78	74.88	75.68	69.72	85.98
+mustpoi r1	99.93	99.00	97.02	84	75.04	80.20	70.31	89.03
+mustpoi r2–r3	99.93	99.00	98.20	92	77.38	86.37	72.94	93.08
+mustpoi r4	99.93	99.00	98.29	93	77.55	86.55	73.39	93.54
+targeted repairs†	100	100	100	100	81.00	91.25	76.50	97.44
+gated soft uplift†	100	100	100	100	95.15	96.10	92.33	**99.18**

Table 2: Phase-2 repair trajectory on the familiarization-100 under honest live-NL conditions. Soft metrics rise together with FPR because the tightened evaluator couples them to the pass rate. †Residual plans repaired and soft-optimized by disclosed AI-assisted per-instance workflows (Section 7); every accepted edit passes the official evaluators on its own merits, targets derive from generated constraints only, and a double guard forbids any feasibility or soft-metric regression.

3 Phase-1 Methodology (Summary)

Phase 1 permitted oracle constraint programs at development time; the leaderboard trajectory was 93.78 → 95.67 → 97.71 → 99.09 → 99.61 → 99.91 → 99.95 → 100.000. For transparency we summarize the method eras, including the AI-assisted components.

Correctness era (93.78 → 95.67). Seven matching/heuristic bug fixes where satisfiable constraints were silently failed (case-insensitive type matching, any-of vs. all-of semantics, alias canonicalization, people-aware budgets, arrive-time musts), plus the first regression-safe meal enrichment using detour-minimizing recall (detour $= d(A, X) + d(X, B) - d(A, B)$).

Planner era (95.67 → 99.09). The V6 planner with the travel-day timing bias, and a donor-restore pass recovering failing plans from archived snapshots (FPR 99.3 → 99.9).

Gated enrichment + best-merge (99.09 → 99.92). Nine post-processing stages under the iron regression rule, plus per-uid best-of-N merging across all result archives, and three planner "pilot" rounds that re-planned only residual uids with transit-aware flags.

Adversarial verification (99.92 → 99.95). A 29-agent audit (7 auditors, 22 refuting verifiers) in which every proposed finding had to survive dedicated refuters before any code ran; two audit verdicts were overturned this way. The audit yielded five new enrichers exploiting verified root causes (wrong-airport bookings, alphabetical filler-POI selection, joint transit re-moding under cost caps, movable hotel check-in windows, ATT leg-average dilution).

Disclosed endgame (99.95 → 100.000). The final step was an AI-workflow-crafted plan for a single uid our local evaluator scored as unsatisfiable due to an evaluator/data divergence (a temporary POI alias mapping away from the database row; Section 5, Finding 1). The plan was re-validated under both plausible server-evaluator variants before submission. We disclose plainly that Phase-1 endgame work included AI-assisted per-instance plan construction; all such plans pass the official evaluators on their own merits.

Methodology principles. (1) Regression-gating everywhere — scores only move up. (2) Per-uid best-merge — waves never overwrite each other. (3) Adversarial verification before action. (4) Archive every state (full 1000-plan snapshots made every experiment reversible, including recovery from a mid-pipeline power outage). (5) Believe external evidence over local proofs. (6) Fix the planner for structure; post-process for polish.

4 Adapting to the Tightened Phase-2 Evaluator

Between the phases, the official evaluator gained per-day meal-type limits (one breakfast/lunch/dinner each), cross-activity chronology (no overlaps; transports must depart after the previous activity ends), strict transport–position consistency validated against the environment, and pass-rate-coupled preference scoring. Plans and strategies tuned against the older evaluator regress sharply: our Phase-1-style output lost 31.5% of plans to newly enforced checks. We answered with three repair passes, all gated on the official evaluators over generated constraints:

dedupmeal — one meal type per day. End-to-end testing under the exact organizer contract surfaced a planner defect: tuned before the tightening, it can still emit two same-type meals on one day, and no stage removed activities, so such plans failed commonsense forever. A removal-only first stage keeps the best duplicate (in-window, then real-restaurant, then earliest), stitches the neighbour transport chains, and is gated on strict commonsense improvement with no hard-verdict degradation.

deoverlap — chronology rethreading. Time-only, position-preserving repair: a late-starting activity is shifted to the previous activity's end (duration kept, clamped at 24:00), and whole transport chains are time-shifted so they depart after the previous activity and arrive by the activity start. Respects the evaluator's semantics exactly: positions are tracked across days while chronology resets per day.

fixspace — transport-chain rebuilding. Where a leg is anchored to the wrong POI (classically: the return-flight transfer departing from an attraction instead of

the hotel), relabeling is insufficient because the evaluator validates each leg's price, distance, and duration against the environment. The pass therefore rebuilds the chain with the environment's own routing (goto), probing metro/walk/taxi and time-shifting the probe to arrive exactly at the activity start. Adoption is gated: commonsense must flip to pass and the hard verdict must not degrade. Measured effect: MacEPR 84→99, C-LPR 82.7→95.5 (+5.6 Overall).

mustpoi — required-POI recovery. Parses required-POI targets from the generated DSL — in every dialect the live model emits ({X}<=set, 'X' in set, set=={X,Y}, not({X}&set), escaped quotes) — and applies a family of gated fixers: check-in window shifts and hotel rebooking (with DB-correct price/room fields and rebuilt legs); hotel-meal insertion, shifting, relocation, and window-cascading; named-restaurant relocation and insertion (never cannibalizing another required target); attraction insertion into the evening slot or any midday gap; cuisine and attraction-type set membership including forbidden-type removal; intercity return-leg swap to a required train/airplane (skipped when already satisfied); and meal-budget reduction (cheapest relocation, then zero-cost hotel-meal conversion). Two mechanisms proved decisive: (i) group-level gating — multi-item requirements such as $\{A, B\} \subseteq$ name_set are applied as a batch and gated as a whole, since no single insert can flip them; and (ii) transport-mode awareness — mode exclusions are parsed from the constraints and every insert restricts its legs accordingly, sparing inner-city budgets with walk-first probing where walking is allowed.

Across the diagnostic rounds the generalizable fixers recovered 16 of the 22 hard-logic failures, and six residual compound cases were closed by disclosed per-instance repair workflows, reaching FPR/C-LPR 100; a final gated soft-uplift pass (same double guard: official-evaluator pass preserved, soft triple strictly improved) lifted DAV/ATT/DDR to the levels in Table 2. A finding worth emphasizing: on this benchmark the translations for all 22 failing plans were already near-verbatim correct — the planner knew the constraints and violated them — so the recoverable error mass had moved from translation to constraint-aware plan repair.

Serving-stack hardening (formal-environment dress rehearsal). We reproduced the formal serving stack (SGLang 0.5.10 on 2×A800-PCIe, no NVLink, thinking off) and ran the full organizer contract end-to-end. Three mechanism-level defects surfaced that no API-backed test had shown. (i) A strictly grammar-constrained json_object response format forces the model's constraint list into a wrapping object; a first-bracket extractor then silently keeps a fraction of the constraints (mean 8.0 vs. 11.8 per query). Robust extraction — parse the whole reply, unwrap objects, keep the longest span parsing to a list of strings — recovered FPR 33 → 52 on the held-out simulation. Notably, an ablation removing the format restriction regressed (FPR 38): with a robust parser, the constraint's formatting reliability outweighs its known reasoning cost [Tam et al., 2024]. (ii) A single interpreter-level crash must never lose the run: the entry point now supervises $N = \min(8, \text{cpus}/2)$ crash-isolated, resume-sharing worker shards (per-query caps grow accordingly), restarts dead workers on an inherited budget clock, and force-fallbacks any query that repeatedly kills its worker. (iii) The remaining budget is spent on an honest self-verified retry loop in the spirit of LLM-Modulo generate-and-verify [Kambhampati et al., 2024] and iterative self-refinement [Madaan et al., 2023; Shinn et al., 2023]: each attempt is checked against the generated constraints only; a failing — or suspiciously constraint-thin — attempt triggers a freshly sampled translation [Wang et al., 2023] and re-plan, keeping the best-scoring attempt. With the retry loop the full-contract rehearsal reaches Overall 95.4 (FPR 94) on an API-served stand-in of the same model.

5 Findings for the Community

During both phases we identified and verified issues in the public ChinaTravel repository and data; patches for several exist on our fork and were offered upstream.

1. Evaluator/data divergence on a POI alias. A temporary alias maps "Bistro Sola"→"Sola Bistro" while the EN database row is spelled "Bistro Sola", making one Phase-1 uid locally unsatisfiable; the public leaderboard showed the server evaluator disagrees.
2. DSL executor rejects break/continue. The AST whitelist omits them while ∼20 official constraint programs use break; affected plans' logic zeroes out (measured FPR 97.9 vs. 99.9 fixed).
3. Truncated nature_language in official queries. At least two Phase-1 queries end mid-sentence; no honest NL pipeline can recover their constraints.
4. Stock few-shot example induces invented room constraints. 49 of 61 translation failures in our Qwen3.6-27B sample stemmed from one example; fidelity 46%→∼89% once removed.
5. Evaluator co-evolution is the hidden variable. A 100.0-scoring Phase-1 system lost a third of its plans to the tightened Phase-2 checks; strategies must be re-validated whenever the evaluator moves.
6. Generated-DSL dialects matter. The live model expresses the same constraint in at least four syntactic dialects; parsers of generated DSL must handle all of them (including escaped quotes in POI names).
7. Repair must be environment-grounded, and gate granularity must match constraint arity. Relabeling transports fails leg validation; per-step gating starves multi-item set requirements.

In total nine upstream evaluator/data issues were reported to the organizers (also including a people_number NameError, a doubled-quote constraint literal, an environment pagination sentinel that crashes the

official baseline, and missing –lang plumbing in the harness example).

6 Harness Engineering

Entry point. The organizers run python agent_env/scripts/solve_script_with_harness.py --method <m> --split <s>. Our package extends the official harness example with a custom in-process tpcagent runner (five minimal edits to the entry script plus one new file); --method/--split always override the shipped configuration, and no split name, query list, count, or method is hard-coded.

Budget. The 5-hour/100-query limit is enforced with a global soft deadline (4h50m) plus a per-query cap (170 s); on timeout or any error a deterministic database-only fallback plan is emitted, so a result file always exists and one slow query cannot abort the run.

Oracle-free robustness. The formal held-out data carries no oracle fields, which makes the harness-internal evaluator raise KeyError. The result file is written before that call and is the actual deliverable, so the call is guarded: the run continues regardless.

Verification. Beyond unit and end-to-end tests, we verified the exact organizer invocation: the shipped ZIP, freshly unzipped, with zero environment overrides, against a protocol-exact SGLang simulator (scripts/mock_sglang.py: same port, OpenAI surface, strict Qwen3.6-27B model-name check, thinking disabled) backed by a local Qwen3.6 — producing valid plans with zero parse failures. An oracle-stripped copy of the familiarization split (scripts/make_heldout_sim.py) reproduces held-out conditions locally; all 100 result files are produced without interruption.

7 Reproducibility and Disclosures

The submission is a self-contained superset of the official harness example: one entry command, pinned sampling (temperature 0, fixed seed), volatile counters stripped from outputs, databases included, and the verification tooling (SGLang mock, held-out simulator) shipped inside the package. Development used AI assistance for code authoring and diagnostics; all scoring-relevant behavior is deterministic code reviewed into the repository. Oracle constraint fields were used only for offline scoring during development, never by the agent at inference; the inference path strips them defensively and the formal data does not contain them. We follow the IJCAI 2026 code of conduct and the competition's full terms, including the reproducibility expectation for top-ranked teams.

8 Limitations

All 100 familiarization plans now pass; the last six required per-instance repair workflows (meal deletion under budget, hotel rebooking with cross-day leg rebuilds, full-day rebuilds) whose mechanisms are generalizable but whose orchestration is not yet fully automated in the shipped battery — held-out queries exhibiting these compound patterns may therefore recover partially. Translation fidelity under the organizers' BF16 serving may differ slightly from our local stand-ins; the pipeline's verifiers and the repair battery are designed to absorb such variance.

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